



Traits & Heredity

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A **trait** is a feature you can observe (eye color, height). **Heredity** is the passing of traits from parents to offspring. You inherited a set of instructions from each parent.

⚠️ **Not all traits are inherited.** A scar, sunburn, or pierced ear are *acquired* during life and can't be passed on.

Genes & Chromosomes

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Your traits are written in **DNA**, organized like nesting dolls:

Cell > Chromosome > Gene > **DNA**

Term	What it is
DNA	The chemical code carrying all instructions for life
Gene	A short section of DNA that controls one trait
Chromosome	A long, coiled strand of DNA; genes sit along it

🔑 Humans have **46 chromosomes (23 pairs)** per body cell — 23 from each parent.

Dominant & Recessive Alleles

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An **allele** is a version of a gene; you have two per trait (one from each parent).

Tongue rolling

T = can roll (dominant) • t = can't roll (recessive)

Combo	Name	Result
TT	Homozygous dominant	Can roll
Tt	Heterozygous	Can roll (T hides t)
tt	Homozygous recessive	Cannot roll

Genotype vs. Phenotype

Genotype = the gene combo (Tt). **Phenotype** = the trait you see ("can roll").

Homozygous = two of the same allele (TT or tt). **Heterozygous** = two different (Tt). A recessive trait shows only as **tt**.

Punnett Squares

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A **Punnett square** predicts the chance of offspring traits. Cross **Tt × Tt**:

Results: **TT 25% • Tt 50% • tt 25%**.

So a **75% chance** of a tongue-roller and 25% chance of a non-roller.

	T	t
T	TT	Tt
t	Tt	tt

★ Top = one parent's alleles, left side = the other. Fill each box by combining its row + column letters.

Asexual vs. Sexual Reproduction

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Feature	Asexual	Sexual
Parents	One	Two
Offspring DNA	Identical (clones)	Unique mix
Genetic variation	None	Lots
Speed	Fast	Slower

🔑 **Why variation matters:** if a disease or change hits, variety means some offspring are likely to survive. Examples of asexual: binary fission (bacteria), budding (yeast), regeneration (starfish), vegetative (strawberry runners).

Selective Breeding

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Humans choose which organisms reproduce so desired traits show up more often. This gave us most pets, farm animals, and crops.

Started with	Bred into
Wild wolves	Hundreds of dog breeds
Wild mustard plant	Broccoli, cabbage, kale, cauliflower
Wild corn (teosinte)	The large sweet corn we eat

 Selective breeding $\neq$ natural selection. In breeding, **humans** pick the traits; in natural selection, the **environment** does.

Quick Check — Heredity

A. Give the genotype name (homozygous dominant / heterozygous / homozygous recessive):

1. Bb $\rightarrow$ _____
2. bb $\rightarrow$ _____
3. BB $\rightarrow$ _____

B. Cross $Tt \times tt$. What % of offspring can roll their tongue? _____ %

C. One parent, identical offspring = _____ reproduction.

I Can... (Self-Check)

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- | | |
|---|---|
| ☐ Tell inherited traits from acquired ones | ☐ Complete a Punnett square & find probability |
| ☐ Order DNA $\rightarrow$ gene $\rightarrow$ chromosome $\rightarrow$ cell | ☐ Compare asexual & sexual reproduction |
| ☐ Use letters for dominant & recessive alleles | ☐ Explain selective breeding |
| ☐ Tell genotype from phenotype | ☐ Explain why genetic variation matters |

 Based on Georgia Standards of Excellence: S7L3 — You're ready for the unit test! 